

13 - ANOVA. Decomposition of Variance. Testing the Equality of Group Means: The F Test. Multivariate ANOVA. Decomposition of Covariance. Testing the Equality of Vector Means: Wilks' Lambda. Bonferroni Confidence Intervals for Treatment Effects.

Applied Statistics

Unat Teksen

March 27, 2026

1 The Big Picture

- **Central question:** Are the means of several groups genuinely different, or are the observed differences just random noise?
- **ANOVA** (Analysis of Variance) answers this for a **single** measured variable per observation.
- **MANOVA** (Multivariate ANOVA) extends the same logic to **multiple** measured variables simultaneously, exploiting correlations between them.
- After rejecting the null hypothesis, **Bonferroni confidence intervals** identify *which* groups differ on *which* variables.
- The core intuition throughout: decompose total variation into *between-group* and *within-group* components, then ask whether the between-group part is surprisingly large relative to the within-group noise.

2 ANOVA — The Model

2.1 Setup and Notation

- $g \geq 2$ groups, n_i observations in group i , total $n = n_1 + \dots + n_g$.
- Indices: $i = 1, \dots, g$ (group), $j = 1, \dots, n_i$ (individual within group).
- **Model equation:**

$$X_{ij} = \mu + \tau_i + \varepsilon_{ij} \tag{1}$$

- X_{ij} : the j -th observation in group i .
- $\mu \in \mathbb{R}$: grand mean (overall average across all groups).
- $\tau_i \in \mathbb{R}$: treatment effect of group i — how much group i shifts observations up or down from μ .
- ε_{ij} : individual random error, pure noise.

- **Error assumption:**

$$\varepsilon_{ij} \sim \text{i.i.d. } N(0, \sigma^2) \quad (2)$$

Errors are independent, normally distributed, and *all groups share the same variance* σ^2 (homoscedasticity).

- **Identifiability constraint:**

$$\sum_{i=1}^g n_i \tau_i = 0 \quad (3)$$

Without this, μ and τ_i cannot be uniquely identified. The constraint pins down the solution by forcing weighted treatment effects to cancel.

2.2 Sample Means

- **Grand sample mean** (unbiased estimator of μ):

$$\bar{X} = \frac{1}{n} \sum_{i=1}^g \sum_{j=1}^{n_i} X_{ij} \quad (4)$$

- **Group sample mean** (estimates $\mu + \tau_i$):

$$\bar{X}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} X_{ij} \quad (5)$$

- The quantity $\bar{X}_i - \bar{X}$ is the natural estimator of τ_i .

3 ANOVA — Hypotheses

- **In terms of group means:**

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_g \quad \text{vs} \quad H_1 : \exists j \neq k \text{ s.t. } \mu_j \neq \mu_k \quad (6)$$

- **Equivalent formulation in terms of treatment effects:**

$$H_0 : \tau_1 = \tau_2 = \dots = \tau_g = 0 \quad \text{vs} \quad H_1 : \exists j \text{ s.t. } \tau_j \neq 0 \quad (7)$$

- The two formulations are identical: all group means equal the grand mean if and only if all treatment effects are zero.
- **Why not just compare means directly?** A difference of 2 cm between group means could arise from the treatment *or* from random noise. ANOVA decides which explanation is more credible.

4 ANOVA — Decomposition of Variance

4.1 Geometric Setup

- Stack all observations into one vector: $\underline{X} = (X_{11}, \dots, X_{1n_1}, X_{21}, \dots, X_{gn_g})^T \in \mathbb{R}^n$.
- For each group i , define the unit vector $\underline{u}_i \in \mathbb{R}^n$ with $1/\sqrt{n_i}$ in positions belonging to group i and 0 elsewhere.
- Key properties of $\underline{u}_1, \dots, \underline{u}_g$:**
 - Linearly independent.
 - Orthogonal: $\underline{u}_j^T \underline{u}_k = 0$ for $j \neq k$.
 - Together they form an *orthonormal basis* for the subspace $\mathcal{L} = \text{Span}(\underline{u}_1, \dots, \underline{u}_g)$.
 - The all-ones vector $\mathbf{1} \in \mathcal{L}$, since $\mathbf{1} = \sum_{i=1}^g \sqrt{n_i} \underline{u}_i$.
- Projection onto $\mathcal{L}$:**

$$\Pi_{\mathcal{L}} \underline{X} = \sum_{i=1}^g \bar{X}_i \underline{u}_i \quad (8)$$

This replaces each observation with its group mean — the vector of fitted values.

- Projection onto $\text{Span}(\mathbf{1})$:**

$$\Pi_{\text{Span}(\mathbf{1})} \underline{X} = \bar{X} \cdot \mathbf{1} \quad (9)$$

This replaces every observation with the grand mean.

4.2 The Three-Way Decomposition

- $\underline{X}$ decomposes into three mutually orthogonal parts:

$$\underline{X} = \underbrace{\bar{X} \cdot \mathbf{1}}_{(1) \text{ grand mean}} + \underbrace{\sum_{i=1}^g (\bar{X}_i - \bar{X}) \underline{u}_i}_{(2) \text{ between-group}} + \underbrace{\left(\underline{X} - \sum_{i=1}^g \bar{X}_i \underline{u}_i \right)}_{(3) \text{ within-group}} \quad (10)$$

- Because these three parts are *mutually orthogonal*, the Pythagorean theorem in $\mathbb{R}^n$ gives:

$$\|\underline{X}\|^2 = \|\bar{X} \cdot \mathbf{1}\|^2 + \left\| \sum_i (\bar{X}_i - \bar{X}) \underline{u}_i \right\|^2 + \left\| \underline{X} - \sum_i \bar{X}_i \underline{u}_i \right\|^2 \quad (11)$$

- Key insight:** The SS decomposition is a Pythagorean theorem in $\mathbb{R}^n$.

4.3 The ANOVA Identity

- Fundamental ANOVA decomposition:**

$$\underbrace{\sum_{i=1}^g \sum_{j=1}^{n_i} (X_{ij} - \bar{X})^2}_{SS_{\text{cent}}} = \underbrace{\sum_{i=1}^g n_i (\bar{X}_i - \bar{X})^2}_{SS_{\text{treat}}} + \underbrace{\sum_{i=1}^g \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2}_{SS_{\text{res}}} \quad (12)$$

- SS_{treat} (between-group): measures how far group means deviate from the grand mean.
- SS_{res} (within-group / residual): measures how far individuals deviate from their own group mean.
- **Intuition:** If treatment has no effect, group means cluster near $\bar{X}$ and SS_{treat} is small relative to SS_{res} . If treatment matters, group means spread apart and SS_{treat} grows large.

5 ANOVA — The F Test

5.1 Distributions of the SS Terms

- **Residual SS:**

$$SS_{\text{res}} \sim \sigma^2 \chi^2(n - g) \quad (13)$$

Degrees of freedom $n - g$: total observations minus the g group means estimated.

- **Treatment SS under H_0 :**

$$SS_{\text{treat}} \sim \sigma^2 \chi^2(g - 1) \quad (14)$$

Degrees of freedom $g - 1$: g group means but one constraint $\sum n_i \tau_i = 0$.

- **Under H_0 :** both $SS_{\text{treat}}/(g - 1)$ and $SS_{\text{res}}/(n - g)$ estimate σ^2 .
- **Under H_1 :** $SS_{\text{treat}}/(g - 1)$ estimates σ^2 *plus* a positive term from treatment effects — it is inflated.
- The three components (1), (2), (3) are **Gaussian** (they are linear/projections of $\underline{X} \sim N_n$) and **independent** (projections onto orthogonal subspaces of a Gaussian vector are independent).

5.2 The F Statistic and Decision Rule

- **F statistic:**

$$F_0 = \frac{SS_{\text{treat}}/(g - 1)}{SS_{\text{res}}/(n - g)} \sim F(g - 1, n - g) \quad \text{under } H_0 \quad (15)$$

- This is the ratio of two independent χ^2 random variables divided by their degrees of freedom.
- **Decision rule:** Reject H_0 at level α if

$$F_0 > F_{1-\alpha}(g - 1, n - g) \quad (16)$$

- **Intuition:** Under H_0 , the ratio ≈ 1 . Under H_1 , the numerator is inflated so the ratio grows large.

5.3 ANOVA Table

Source	SS	df	MS	F
Between groups	SS_{treat}	$g - 1$	$SS_{\text{treat}}/(g - 1)$	$MS_{\text{treat}}/MS_{\text{res}}$
Within groups	SS_{res}	$n - g$	$SS_{\text{res}}/(n - g)$	
Total (centered)	SS_{cent}	$n - 1$		

6 MANOVA — The Model

6.1 Why Not Run Separate ANOVAs?

- In ANOVA, each observation is a *single number*. In practice, you often measure p variables simultaneously on each subject.
- **Example:** Students in 3 teaching methods, each measured on math *and* reading scores.
- Running p separate ANOVAs ignores correlations between variables and inflates the Type I error rate. If each test has error rate α , running p tests gives cumulative error $\approx 1 - (1 - \alpha)^p$.
- MANOVA tests all p variables *simultaneously*, exploiting their joint distribution.

6.2 Setup and Notation

- $p \geq 1$ variables, $g \geq 2$ groups.
- **Model equation:**

$$\underline{X}_{ij} = \underline{\mu} + \underline{\tau}_i + \underline{\varepsilon}_{ij} \quad (17)$$

- $\underline{X}_{ij} \in \mathbb{R}^p$: observation vector.
- $\underline{\mu} \in \mathbb{R}^p$: grand mean vector.
- $\underline{\tau}_i \in \mathbb{R}^p$: treatment effect vector for group i .
- $\underline{\varepsilon}_{ij}$: error vector.

- **Error assumption:**

$$\underline{\varepsilon}_{ij} \sim \text{i.i.d. } N_p(\underline{0}, \Sigma) \quad (18)$$

Σ is a $p \times p$ covariance matrix, *shared by all groups* (multivariate homoscedasticity).

- **Identifiability constraint:**

$$\sum_{i=1}^g n_i \underline{\tau}_i = \underline{0} \quad (19)$$

- **Component-wise:** For each variable $k = 1, \dots, p$:

$$X_{ijk} = \mu_k + \tau_{ik} + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \sim N(0, \sigma_{kk}) \quad (20)$$

where σ_{kk} is the k -th diagonal entry of Σ . MANOVA for component k is ANOVA on the k -th variable. MANOVA runs all of these simultaneously using the full correlation structure.

- **Covariance structure:**

$$\Sigma = \begin{pmatrix} \sigma_{11} & \cdots & \sigma_{1p} \\ \vdots & \ddots & \vdots \\ \sigma_{p1} & \cdots & \sigma_{pp} \end{pmatrix} \quad (21)$$

Diagonal entries σ_{kk} : variance of variable k . Off-diagonal σ_{kl} : covariance between variables k and l .

7 MANOVA — Hypotheses and Decomposition of Covariance

7.1 Hypotheses

- **In terms of treatment vectors:**

$$H_0 : \tau_1 = \tau_2 = \cdots = \tau_g = \underline{0} \quad \text{vs} \quad H_1 : \exists j \text{ s.t. } \tau_j \neq \underline{0} \quad (22)$$

- **Equivalent formulation:**

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_g \quad \text{vs} \quad H_1 : \text{at least one pair differs} \quad (23)$$

7.2 Decomposition of Covariance

- ANOVA decomposed a scalar SS. MANOVA decomposes a $p \times p$ scatter matrix:

$$\underbrace{\sum_{i=1}^g \sum_{j=1}^{n_i} (\underline{X}_{ij} - \bar{\underline{X}})(\underline{X}_{ij} - \bar{\underline{X}})^T}_{\text{Total scatter}} = \underbrace{\sum_{i=1}^g n_i (\bar{\underline{X}}_i - \bar{\underline{X}})(\bar{\underline{X}}_i - \bar{\underline{X}})^T}_B + \underbrace{\sum_{i=1}^g \sum_{j=1}^{n_i} (\underline{X}_{ij} - \bar{\underline{X}}_i)(\underline{X}_{ij} - \bar{\underline{X}}_i)^T}_W \quad (24)$$

- **Why outer products?** In the scalar case, $(x - \bar{x})^2$ measures squared deviation. For vectors, $(\underline{x} - \bar{\underline{x}})(\underline{x} - \bar{\underline{x}})^T$ is a $p \times p$ matrix capturing both squared deviations in each direction *and* co-deviations across directions.
- **Proof hint:** Start from the identity $\underline{X}_{ij} - \bar{\underline{X}} = (\underline{X}_{ij} - \bar{\underline{X}}_i) + (\bar{\underline{X}}_i - \bar{\underline{X}})$, expand the outer product, sum over i, j ; the cross term vanishes by orthogonality.
- **Between-group scatter matrix B :**

$$B = \sum_{i=1}^g n_i (\bar{\underline{X}}_i - \bar{\underline{X}})(\bar{\underline{X}}_i - \bar{\underline{X}})^T \quad (25)$$

Diagonal entries of B : SS_{treat} for each variable separately.

- **Within-group scatter matrix W :**

$$W = \sum_{i=1}^g \sum_{j=1}^{n_i} (\underline{X}_{ij} - \bar{\underline{X}}_i)(\underline{X}_{ij} - \bar{\underline{X}}_i)^T \quad (26)$$

Diagonal entries of W : SS_{res} for each variable separately.

- **Reject H_0 if B/W is “large.”** But B and W are matrices — we need a scalar summary of their relative size.

8 MANOVA — Test Statistics

8.1 Three Proposals

- **1. Wilks' Lambda (most common):**

$$\Lambda_W = \frac{\det(W)}{\det(B + W)} \quad (27)$$

- $\det$ (determinant) measures the “volume” of a matrix — the generalized variance.
- $\Lambda_W = 1$: all variation is within groups, no treatment effect.
- $\Lambda_W = 0$: all variation is between groups, perfect separation.
- **Reject H_0 if Λ_W is small.**

- **2. Pillai's Lambda:**

$$\Lambda_P = \text{tr}(B(B + W)^{-1}) \quad (28)$$

- tr (trace) = sum of diagonal entries.
- Measures the proportion of total variation explained by treatment in each direction, summed over all directions.
- Most robust statistic — performs well even when assumptions are mildly violated.
- **Reject H_0 if Λ_P is large.**

- **3. Hotelling-Lawley Lambda:**

$$\Lambda_{HL} = \text{tr}(BW^{-1}) \quad (29)$$

- The direct matrix analog of the scalar ANOVA F-statistic.
- **Reject H_0 if Λ_{HL} is large.**

8.2 Summary Table

Statistic	Formula	Reject H_0 if
Wilks Λ_W	$\det(W)/\det(B + W)$	small
Pillai Λ_P	$\text{tr}(B(B + W)^{-1})$	large
Hotelling-Lawley Λ_{HL}	$\text{tr}(BW^{-1})$	large

- When $g = 2$ (two groups), all three statistics give identical decisions.
- For $g > 2$ they may differ slightly.

9 MANOVA — Distribution of Wilks' Lambda and the Test

9.1 Exact Distribution

- The exact null distribution of Λ_W is known only in special cases:

- $p \geq 1, g = 2$ or $g = 3$: exact distribution known (reduces to an F-distribution).
- $p = 2, g \geq 1$: exact distribution known.
- General p, g : exact distribution *unknown* — approximations are needed.

9.2 Bartlett's Approximation (Large n)

- For large n (as $n \rightarrow \infty$), Bartlett showed:

$$-\left(n - 1 - \frac{p+g}{2}\right) \log \Lambda_W \sim \chi^2(p(g-1)) \quad (30)$$

- The log transform converts the multiplicative structure of Λ_W into an additive one that approximates χ^2 .
- Degrees of freedom $p(g-1)$: $g-1$ free treatment effect vectors (due to the constraint), each in $\mathbb{R}^p$.
- The factor $(n - 1 - \frac{p+g}{2})$ is a finite-sample correction improving the approximation.
- **Decision rule:** Reject H_0 at level α if

$$-\left(n - 1 - \frac{p+g}{2}\right) \log \Lambda_W \geq \chi^2_{1-\alpha}(p(g-1)) \quad (31)$$

9.3 Independence and Gaussianity of Components

- Components (1), (2), (3) of the decomposition are **Gaussian**: they are projections (linear maps) of $\underline{X} \sim N_n$, so they inherit Gaussianity.
- They are **independent**: projections of a Gaussian vector onto *orthogonal* subspaces are always independent. The joint distribution of $P\underline{X}$ and $(I - P)\underline{X}$ is multivariate normal with block-diagonal covariance (zero off-diagonal block).
- This independence is essential — it ensures B and W follow (scaled) Wishart distributions and allows the F / χ^2 approximations.

10 Bonferroni Confidence Intervals for Treatment Effects

10.1 Motivation

- Rejecting H_0 in MANOVA only says: “at least one group differs on at least one variable.”
- To identify *which* groups differ on *which* variables, we construct **Bonferroni confidence intervals**.
- We want CIs for $\tau_{ik} - \tau_{jk}$: the difference in treatment effect between groups i and j on variable k .
- Indices: $i, j \in \{1, \dots, g\}$ (group pairs), $k \in \{1, \dots, p\}$ (variables).

10.2 The Multiple Comparison Problem

- Total number of simultaneous CIs:

$$c = p \cdot \binom{g}{2} = p \cdot \frac{g(g-1)}{2} \quad (32)$$

- **Example:** $g = 3$ groups, $p = 2$ variables $\Rightarrow c = 2 \times 3 = 6$ CIs.
- If each CI has individual error rate α , the probability of at least one error across c CIs is approximately $1 - (1 - \alpha)^c$.
- **Bonferroni correction:** Use individual level α/c so that the simultaneous coverage is at least $1 - \alpha$.

10.3 The Estimator and Its Distribution

- **Natural estimator:**

$$\widehat{\tau_{ik} - \tau_{jk}} = \bar{X}_{ik} - \bar{X}_{jk} \quad (33)$$

- **Sampling distribution:**

$$\bar{X}_{ik} - \bar{X}_{jk} \sim N\left(\tau_{ik} - \tau_{jk}, \sigma_{kk}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)\right) \quad (34)$$

because groups are independent and each group mean has variance σ_{kk}/n_i .

- **Pooled estimator of σ_{kk} :**

$$\frac{W_{kk}}{n - g} = \frac{1}{n - g} \sum_{i=1}^g (n_i - 1) S_{i,kk} \quad (35)$$

where $S_{i,kk}$ is the sample variance of variable k in group i , and

$$S_i = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (\underline{X}_{ij} - \bar{X}_i)(\underline{X}_{ij} - \bar{X}_i)^T \quad (36)$$

Pooling combines variance information across all groups for a more precise estimate.

10.4 The Bonferroni CI Formula

- **Bonferroni CI at simultaneous level $1 - \alpha$:**

$$\text{Bonf CI}_{1-\alpha}(\tau_{ik} - \tau_{jk}) = \left[\bar{X}_{ik} - \bar{X}_{jk} \pm t_{1-\frac{\alpha}{2c}}(n - g) \sqrt{\frac{W_{kk}}{n - g} \left(\frac{1}{n_i} + \frac{1}{n_j}\right)} \right] \quad (37)$$

where $c = p \cdot \frac{g(g-1)}{2}$.

- **Components explained:**

- $\bar{X}_{ik} - \bar{X}_{jk}$: center of the CI, the observed difference in group means.
- $t_{1-\alpha/(2c)}(n - g)$: Bonferroni-corrected t -quantile; grows larger as c increases, widening the intervals.

- $W_{kk}/(n - g)$: pooled variance estimate of σ_{kk} .
- $\sqrt{1/n_i + 1/n_j}$: standard error factor; larger groups yield narrower intervals.
- We use a t -distribution (not normal) because σ_{kk} is estimated. Degrees of freedom $n - g$: W uses g estimated group means.
- **Interpretation:** If the CI for $\tau_{ik} - \tau_{jk}$ does *not* contain 0, the two groups differ significantly on variable k .

10.5 Complete Correspondence: ANOVA vs MANOVA

Concept	ANOVA	MANOVA
Observation	$X_{ij} \in \mathbb{R}$	$\underline{X}_{ij} \in \mathbb{R}^p$
Treatment effect	$\tau_i \in \mathbb{R}$	$\underline{\tau}_i \in \mathbb{R}^p$
Error distribution	$N(0, \sigma^2)$	$N_p(\underline{0}, \Sigma)$
Between-group	SS_{treat} (scalar)	B ($p \times p$ matrix)
Within-group	SS_{res} (scalar)	W ($p \times p$ matrix)
Test statistic	F ratio	$\Lambda_W, \Lambda_P, \Lambda_{HL}$
Post-hoc CIs	t -based CIs	Bonferroni CIs with t_{n-g}

Türkçe Açıklamalar

1. Büyük Resim

- **Temel soru:** Birden fazla grubun ortalaması gerçekten farklı mı, yoksa bu farklar sadece şans eseri mi?
- **ANOVA:** Her gözlem *tek* bir sayıysa kullanılır.
- **MANOVA:** Her gözlem *birden fazla* sayıdan oluşan bir vektörse kullanılır; değişkenler arasındaki korelasyonu da hesaba katar.
- **Bonferroni güven aralıkları:** H_0 reddedildikten sonra hangi grupların hangi değışkonde farklı olduğunu belirler.
- **Temel sezgi:** Toplam varyasyonu *gruplar arası* ve *gruplar içi* olarak ikiye böl; gruplar arası parçanın gürültüye kıyasla büyük olup olmadığını sor.

2. ANOVA — Model

- $g \geq 2$ grup, grup i 'de n_i gözlem, toplam $n = n_1 + \dots + n_g$.
- **Model denklemi:** $X_{ij} = \mu + \tau_i + \varepsilon_{ij}$
 - μ : genel ortalama (tüm gözlemlerin ortalaması).
 - τ_i : grup i 'nin tedavi etkisi — genel ortalamadan ne kadar sapılıyor?
 - ε_{ij} : bireysel rastlantı hatası.
- **Hata varsayımı:** $\varepsilon_{ij} \sim \text{i.i.d. } N(0, \sigma^2)$ — tüm gruplar aynı σ^2 varyansını paylaşır (homojenlik varsayımı).
- **Tanımlanabilirlik kısıtı:** $\sum_{i=1}^g n_i \tau_i = 0$ — Bu olmadan μ ve τ_i 'ler tek başına tanımlanamaz; kısıt çözümü sabitler.
- **Genel ortalama** $\bar{X}$, μ 'nun tarafsız tahmincisidir. **Grup ortalaması** $\bar{X}_i$, $\mu + \tau_i$ 'yi tahmin eder. $\bar{X}_i - \bar{X}$, τ_i 'nin doğal tahmincisidir.

3. ANOVA — Hipotezler

- Grup ortalamaları cinsinden: $H_0 : \mu_1 = \dots = \mu_g$ vs H_1 : en az bir çift farklı.
- Eşdeğer olarak tedavi etkileri cinsinden: $H_0 : \tau_1 = \dots = \tau_g = 0$ vs $H_1 : \exists j, \tau_j \neq 0$.
- **Neden sadece ortalamalara bakamayız?** Örneğin 2 cm'lik fark tedaviden mi gürültüden mi kaynaklanıyor? ANOVA bu ayrımı yapar.

4. ANOVA — Varyans Ayrışması

- Tüm veriler tek bir $\underline{X} \in \mathbb{R}^n$ vektörüne yığılır. Her grup için birim vektör $\underline{u}_i$ tanımlanır. $\underline{u}_i$ 'ler birbirine dik ve birim uzunluktadır; $\mathcal{L} = \text{Span}(\underline{u}_1, \dots, \underline{u}_g)$ altuzayını oluşturur.
- $\underline{X}$ üç dik parçaya ayrılır: genel ortalama bileşeni (1) + gruplar arası bileşen (2) + gruplar içi bileşen (3).

- **Pisagor teoremi:** Bu parçalar dik olduğundan $\|\underline{X}\|^2$ üç parçanın ℓ^2 normlarının toplamına eşittir. Bu varyans ayrışmasının geometrik özüdür.
- **Temel ANOVA özdeşliği:**

$$SS_{\text{cent}} = SS_{\text{treat}} + SS_{\text{res}}$$

- $SS_{\text{treat}} = \sum_i n_i (\bar{X}_i - \bar{X})^2$: grup ortalamaları genel ortalamadan ne kadar uzak?
- $SS_{\text{res}} = \sum_i \sum_j (X_{ij} - \bar{X}_i)^2$: bireyler kendi grup ortalamasından ne kadar uzak?
- **Sezgi:** Tedavi etkisi yoksa grup ortalamaları $\bar{X}$ 'e yakın $\rightarrow SS_{\text{treat}}$ küçük. Tedavi etkisi varsa grup ortalamaları yayılır $\rightarrow SS_{\text{treat}}$ büyür.

5. ANOVA — F Testi

- $SS_{\text{res}} \sim \sigma^2 \chi^2(n - g)$; $SS_{\text{treat}} \sim \sigma^2 \chi^2(g - 1)$ (H_0 altında).
- H_0 altında her iki parça da σ^2 'yi tahmin eder. H_1 altında $SS_{\text{treat}}/(g - 1)$ şişer.
- **F istatistiği:** $F_0 = \frac{SS_{\text{treat}}/(g - 1)}{SS_{\text{res}}/(n - g)} \sim F(g - 1, n - g)$
- **Karar:** $F_0 > F_{1-\alpha}(g - 1, n - g)$ ise H_0 'ı reddet.
- **Sezgi:** H_0 altında oran ≈ 1 . H_1 altında pay şişer, oran büyür.

6. MANOVA — Model

- **Neden ayrı ANOVA'lar yetmez?** Değişkenler arasındaki korelasyon göz ardı edilir ve toplam hata oranı artar: p test için kümülatif hata $\approx 1 - (1 - \alpha)^p$.
- **Model:** $\underline{X}_{ij} = \underline{\mu} + \underline{\tau}_i + \underline{\varepsilon}_{ij}$, her şey vektör.
- **Hata varsayımı:** $\underline{\varepsilon}_{ij} \sim \text{i.i.d. } N_p(\underline{0}, \Sigma)$. Σ tüm gruplar için ortak $p \times p$ kovaryans matrisi.
- **Bileşen bazında:** Her k bileşeni için MANOVA, k . değişkende ANOVA'ya indirgenebilir; MANOVA tüm bu ANOVA'ları korelasyonları kullanarak eş zamanlı yapar.

7. MANOVA — Kovaryans Ayrışması ve Hipotezler

- **Hipotezler:** $H_0 : \underline{\mu}_1 = \dots = \underline{\mu}_g$ (tüm grup ortalama vektörleri eşit) vs H_1 : en az biri farklı.
- ANOVA'da SS sayırken, MANOVA'da $p \times p$ matrisler:
 - B : gruplar arası saçılma matrisi. Köşegenleri her değişken için SS_{treat} .
 - W : gruplar içi saçılma matrisi. Köşegenleri her değişken için SS_{res} .
 - $B + W$: toplam saçılma matrisi.
- Dış çarpım $(\underline{x})(\underline{x})^T$ kullanılır çünkü hem her yöndeki saçılmayı hem de yönler arası ilişkiyi (kovaryansı) yakalar.
- H_0 'ı reddet eğer B/W “büyükse” — ama matris oranını nasıl ölçeceğiz?

8. MANOVA — Test İstatistikleri

- **Wilks' Lambda:** $\Lambda_W = \det(W) / \det(B+W)$. $\det$ (determinant) matrisin kapladığı “hacmi” ölçer. $\Lambda_W = 1 \rightarrow$ etki yok, $\Lambda_W = 0 \rightarrow$ mükemmel ayırım. **Küçükse reddet.**
- **Pillai's Lambda:** $\Lambda_P = \text{tr}(B(B+W)^{-1})$. İz = köşegen elemanlar toplamı. Varsayımlar biraz bozulsan bile en sağlam istatistik. **Büyükse reddet.**
- **Hotelling-Lawley:** $\Lambda_{HL} = \text{tr}(BW^{-1})$. F istatistiğinin matris versiyonu. **Büyükse reddet.**
- $g = 2$ için üçü aynı kararı verir. $g > 2$ için hafif farklılık olabilir.

9. Wilks' Lambda Dağılımı ve Bartlett Yaklaşımı

- Kesin dağılım yalnızca $g = 2, 3$ veya $p = 2$ durumlarında bilinir. Genel durumda bilinmez.
- **Bartlett yaklaşımı (büyük n):**

$$-(n-1 - \frac{p+g}{2}) \log \Lambda_W \sim \chi^2(p(g-1))$$

$\log$ dönüşümü çarpımsal yapıyı toplamsal hale getirir. Serbestlik derecesi $p(g-1)$: $g-1$ serbest tedavi etki vektörü, her biri p boyutlu.

- **Karar:** Test istatistiği $\chi^2_{1-\alpha}(p(g-1))$ 'den büyükse H_0 'ı reddet.
- **Bileşenlerin Gausslığı ve bağımsızlığı:** Projeksiyon operatörleri doğrusal olduğundan bileşenler Gaussian'dır. Dik altuzaylara yapılan projeksiyonlar Gaussian vektörler için her zaman bağımsızdır; bu B ve W 'nin χ^2 / Wishart dağılımlarını takip etmesini sağlar.

10. Bonferroni Güven Aralıkları

- **Amaç:** MANOVA H_0 'ı reddettikten sonra hangi grup çifti, hangi değişkende farklı?
- **Çoklu karşılaştırma problemi:** Toplam CI sayısı $c = p \cdot \frac{g(g-1)}{2}$. Örnek: $g = 3$, $p = 2$ için $c = 6$. Her birinde ayrı ayrı α hata oranı kullanılsaydı kümülatif hata $\approx 1 - (0.95)^6 \approx 0.26$ olurdu. Bonferroni: her CI'de α/c kullan.
- **Tahmin edici:** $\widehat{\tau_{ik} - \tau_{jk}} = \bar{X}_{ik} - \bar{X}_{jk}$, dağılımı $N(\tau_{ik} - \tau_{jk}, \sigma_{kk}(1/n_i + 1/n_j))$.
- **Pooled varyans:** $W_{kk}/(n-g)$ — tüm grupların varyans bilgisi bir havuzda birleştirilir, daha iyi tahmin sağlar.
- **Bonferroni CI formülü:**

$$\left[\bar{X}_{ik} - \bar{X}_{jk} \pm t_{1-\frac{\alpha}{2c}}(n-g) \sqrt{\frac{W_{kk}}{n-g} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)} \right]$$

- t dağılımı kullanılır çünkü σ_{kk} tahmin edilmiştir (bilinmiyor).
- Serbestlik derecesi $n-g$: W hesaplanırken g grup ortalaması kullanıldı.

- CI **0 içermiyorsa** $\rightarrow$ o grup çifti, o değişkende istatistiksel olarak anlamlı biçimde farklı.
- c arttıkça t quantili büyür $\rightarrow$ aralıklar genişler $\rightarrow$ daha muhafazakâr.